

ILE ROUGE, QC, Canada

WMO: 714260

Lat: 48.069N

Lon: 69.556W

Elev: 19

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.7	-1.5	-11.0	3.0	-4.9	-7.3	3.7	-0.4	49.1	7.2	45.4	9.8	24.7	290	0.890

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	64.6	59.0	62.5	57.7	60.8	56.5	60.6	63.2	58.9	61.5	57.4	60.1	11.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	75.4	61.9	57.5	70.7	60.3	56.0	66.9	58.9	26.8	63.4	25.7	61.6	24.7	60.2	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
39.6	35.0	30.7	-10.5	71.7	3.4	3.4	-13.0	74.1	-15.0	76.0	-17.0	77.9	-19.4	80.4
			-11.8	65.4	3.4	2.9	-14.2	67.5	-16.2	69.2	-18.1	70.8	-20.5	72.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	15.8	17.7	26.1	35.6	44.4	51.4	55.1	54.6	49.7	42.9	34.0	23.2
	DBStd	15.04	10.20	9.28	8.03	5.09	4.00	3.89	3.49	3.39	3.91	4.35	6.46	8.67
	HDD50	4944	1061	906	740	432	181	28	4	4	52	224	481	831
	HDD65	9986	1526	1326	1205	882	640	407	308	321	459	685	931	1296
	CDD50	430	0	0	0	0	6	71	160	148	42	3	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	14.8	19.8	19.2	17.4	14.6	12.1	11.6	11.0	11.0	12.3	14.2	16.3	18.6
(15) Precipitation	PrecAvg	40.6	2.7	2.6	2.6	3.2	3.3	3.6	4.4	4.2	3.9	4.1	3.2	3.4
	PrecMax	49.1	4.8	4.4	5.0	7.6	6.2	7.3	10.6	7.9	10.3	8.7	5.0	9.5
	PrecMin	31.7	0.2	1.4	1.2	0.6	1.2	1.4	2.1	1.9	2.0	1.3	1.7	1.0
	PrecStd	4.9	1.2	0.8	1.0	1.6	1.3	1.5	2.2	1.9	1.9	1.8	0.9	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	37.4	43.6	49.2	59.3	65.6	68.3	67.9	63.3	56.3	52.6	41.1
		MCWB	37.5	35.3	40.5	41.8	51.3	57.6	61.3	59.6	58.0	51.3	50.6	38.9
	2%	DB	35.1	34.5	38.6	45.6	55.5	62.0	64.8	64.2	59.9	53.3	46.1	37.8
		MCWB	33.8	32.9	35.5	40.4	49.2	56.1	59.7	58.7	56.1	49.7	43.8	35.9
	5%	DB	32.6	32.4	36.5	43.2	52.7	59.8	62.7	62.1	57.5	51.1	43.6	36.1
		MCWB	31.4	31.1	34.0	39.1	47.6	54.6	58.3	57.4	54.2	47.9	41.1	34.5
	10%	DB	29.7	30.7	34.8	41.4	50.4	57.6	60.9	60.2	55.5	49.0	41.6	34.3
		MCWB	28.2	29.5	32.7	37.8	46.1	53.1	56.9	56.3	52.5	46.2	39.2	32.8
	0.4%	WB	37.8	36.2	38.0	43.4	53.2	60.7	63.6	62.5	59.9	52.9	51.1	39.3
		MCDB	38.2	37.1	39.7	46.5	56.2	63.2	66.5	65.6	61.7	54.9	52.9	40.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.2	33.4	35.5	41.1	50.3	58.0	61.0	60.1	56.7	50.3	44.8	36.4
		MCDB	35.0	34.3	36.8	44.1	53.5	60.7	63.6	63.0	58.8	52.4	46.0	37.4
	5%	WB	31.6	31.4	34.0	39.6	48.1	55.8	59.3	58.4	54.8	48.5	41.7	34.8
		MCDB	32.6	32.4	35.1	42.4	51.2	58.5	61.9	61.1	56.9	50.9	43.2	35.9
	10%	WB	28.5	29.4	32.7	38.3	46.2	54.0	57.8	56.9	52.9	46.7	39.6	32.8
		MCDB	29.8	30.4	34.0	40.8	49.3	56.9	60.4	59.7	55.0	49.2	41.1	34.0
	5% DB	MDBR	10.6	9.9	8.5	7.6	8.8	10.7	11.8	11.5	9.3	6.4	6.3	8.2
		MCDBR	12.4	8.6	8.6	10.2	13.1	14.3	14.2	14.3	12.0	9.3	7.8	8.8
		MCWBR	12.8	8.4	7.4	7.2	8.7	9.8	10.1	10.1	8.9	7.4	7.7	8.0
		MCWBR	12.5	8.6	7.5	9.4	12.7	13.8	13.5	13.7	11.6	9.3	7.9	8.1
(40) Clear-Sky Solar Irradiance	taub		0.270	0.280	0.318	0.360	0.369	0.387	0.414	0.394	0.370	0.334	0.299	0.276
		taud	2.326	2.288	2.247	2.283	2.345	2.360	2.310	2.370	2.426	2.504	2.460	2.352
	Ebn at Noon		257	280	284	282	283	278	268	269	265	259	245	238
		Edh at Noon	22	29	36	39	38	38	39	35	30	24	20	19
(44) All-Sky Solar Radiation	RadAvg		410	702	1075	1401	1622	1738	1721	1483	1144	670	365	277
	RadStd		46	81	80	117	159	121	168	98	79	57	39	26

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page